Quantum Entanglement: From the EPR Paradox to Experimental Reality

This poster reviews quantum entanglement from the EPR paradox and Bell's theorem to experimental verification. It's technological applications, methodical limitations, and future directions in quantum computing, communication, and sensing.

Aumrika Joarder
Department of Electronics and Communication Engineering
Bharati Vidyapeeth's College of Engineering, New Delhi

01 Quantum Entanglement

The phenomenon in which the quantum state of each particle in a group cannot be described independently of the state of the others, even when the particles are separated by a large distance.

02 EPR Pradox

The EPR paradox is a thought experiment proposed in 1935 by Albert Einstein, Boris Podolsky, and Nathan Rosen to challenge the completeness of quantum mechanics, arguing that quantum entanglement suggests correlations between distant particles that appear to conflict with the principle of local realism and cannot fully describe reality because measuring one property limits knowledge of another.

03 Bell's Theorem

Bell's theorem proves that quantum physics is incompatible with "local realism." It shows that no hidden local variables can explain the instantaneous, entangled connections observed between distant particles.

Bell's Inequality
Bell's inequality is a mathematical formula from physics that proves quantum mechanics cannot be explained by classical local hidden-variable theories.
$|S| \leq 2$ (CHSH Form)

04 Experimental Proofs

Freedman and Clauser Experiment (1972) — 1972
Their measurements of calcium atomic cascades strongly supported quantum mechanics. Using highly accurate data, they violated Bell's inequality, providing decisive evidence against local hidden-variable theories.

Aspect and colleagues' experiments (1982) — 1982
They proved that entangled photons violate Bell's inequalities. This conclusively rejected local realism, confirming quantum mechanics' non-local nature and proving Einstein's "spooky action at a distance" is real.

Anton Zeilinger's Group (1998) — 1998
They pioneered quantum teleportation (1997) and advanced entanglement swapping, laying the foundation for secure quantum communication networks and quantum computing.

Loophole-Free Bell Tests (2015) — 2015
Confirmed quantum entanglement by violating Bell's inequality, definitively proving quantum nonlocality and ruling out local hidden-variable theories.

05 Technological Applications

Quantum Computing: Uses qubits tied together via entanglement to enable computational approaches that can outperform classical methods for specific problems.
Quantum Sensing and Metrology: Uses entangled particles to sharply boost measurement precision for time and gravity.
Quantum Imaging: Uses entangled-photon correlations for specialized imaging techniques, including applications in low-light imaging and quantum-enhanced measurements.
Quantum Cryptography: Uses quantum principles, particularly in quantum key distribution (QKD), to establish secure encryption keys and detect potential eavesdropping.

06 Open Questions and Challenges

Scalability: How can entanglement be scaled to large quantum systems while maintaining reliability?
Decoherence: How can entangled states be protected from environmental disturbances?
Efficient Measurement: How can entanglement be measured accurately with fewer resources?
Bound Entanglement: Why can some entangled states not be used for standard quantum information tasks?
Macroscopic Entanglement and Network Integration: How can entanglement be maintained in large systems and distributed reliably across quantum networks?
Limits of Quantum Nonlocality: Why does quantum mechanics permit nonlocal correlations but limit how strong they can be?

References
Can Quantum-Mechanical Description of Physical Reality Be Considered Complete? - Albert Einstein, Boris Podolsky, and Nathan Rosen
On the Einstein Podolsky Rosen Paradox. - John S. Bell
50 years of Experimental Bell inequalities: a short introduction - Alessio Avella, Marco Genovese
The quantum technologies roadmap: a European community view - Antonio Acín et al 2018 New J. Phys. 20 080201